Hydroponics Monitoring System

SysML diagrams and integrity review | Model 1.0 | 16 October 2026

This package defines one connected architecture model for the redesigned monitoring system. It contains 15 diagram sheets covering all seven requested diagram types, with stable requirement IDs, typed internal interfaces, and behavior tied to the current power and firmware design.

Model boundary and baseline

The system contains one ESP32-S3 field node, the Python BLE commissioning application, and the monitoring backend/dashboard. External power, the measured hydroponic environment, the laptop platform, and the controlled load form its context.

The baseline is the current Redesign Specification Rev 5, including the compact voltage notation and uppercase BLE update. Hardware assignments follow the current pin and power drawings. The older firmware architecture contributes the ownership, scheduling and recovery concepts, adapted to ESP32-S3 and SGP30.

View	Sheets	Diagram IDs
Requirement diagram	3	REQ-01 to REQ-03
Block definition diagram	3	BDD-01 to BDD-03
Internal block diagram	2	IBD-01 and IBD-02
Activity diagram	2	ACT-01 and ACT-02
Sequence diagram	3	SD-01 to SD-03
State machine diagram	1	STM-01
Use case diagram	1	UC-01

Reading the diagrams

BDD boxes define types; IBD labels identify part usages as name : Type. Filled diamonds mark the composite whole. Dashed dependency arrows point to their supplier, including the source requirement or included use case. Proxy ports expose the logical interface contracts detailed in BDD-03 and the interface register.

Verification status

All 18 implementation requirements have design-responsibility and planned-test links. The shared model passes the recorded structural, control-flow, typing and bounded protocol checks. These checks verify the diagram model; the ten electrical and firmware test cases remain specified for execution on the prototype.

Diagram sheets use A3 landscape for readable engineering notation. Registers and review pages use A4. The PDF bookmarks navigate directly to each view.

REQ-01 Functional requirements and project goals

req [package Requirements]

All dashed dependencies in this view are «deriveReqt» relationships. Arrowheads point from the derived requirement to its source.

«requirement»
Hydroponics monitoring system

id = "R0"
text = "Deliver integrated sensing, local control, outage continuity, BLE commissioning and remote monitoring with traceable validation."

«deriveReqt» «deriveReqt» «deriveReqt» «deriveReqt»

«requirement»
Integrated controller

id = "G1"
text = "Deliver the custom ESP32-S3 controller PCB."

«requirement»
Qualified measurements

id = "G2"
text = "Integrate and qualify the seven sensor functions."

«requirement»
Local commissioning

id = "G5"
text = "Configure and maintain the controller through the Python BLE application."

«requirement»
Remote monitoring

id = "G6"
text = "Provide live data, history, alarms, and node health."

«requirement»
Controller implementation

id = "R01"
text = "Use ESP32-S3-WROOM-1-N16R8 on the custom PCB, with the current power and sensor interfaces."

«requirement»
Seven sensor functions

id = "R03"
text = "Acquire BME280, DS18B20, DFR0026, SGP30, HC-SR04, pH-4502C, and SEN0244 data."

«requirement»
Secure BLE access

id = "R11"
text = "Use authenticated encrypted BLE, deliberate commissioning access, write-only credentials, and explicit request results."

«requirement»
Dashboard information

id = "R14"
text = "Present live data, history, alarms, quality, age, calibration status, connectivity and power mode."

«requirement»
Board interface contract

id = "R02"
text = "Implement the fixed GPIO map, protected signal interfaces, USB access, reset and boot controls."

«requirement»
Measurement qualification

id = "R04"
text = "Attach units, age, quality and calibration revision; reject invalid compensation and preserve last-good data separately."

«requirement»
Configuration transactions

id = "R12"
text = "Validate, stage, persist a versioned configuration, and apply at a safe boundary; recover the last valid commit."

«requirement»
Network fault tolerance

id = "R15"
text = "Continue local sensing, BLE and local rules during network loss; use bounded reconnects and offline history."

«requirement»
Supported device management

id = "R13"
text = "Accept only supported types and board slots; reject conflicts, duplicate identities and invalid ranges without reflashing."

Derivation arrows point from the derived requirement to its source. Requirements recur in other views by the same ID. Full satisfaction and verification coverage is in the trace matrix.

REQ-02 Continuity and local behavior requirements

req [package Requirements]

All dashed dependencies in this view are «deriveReqt» relationships. Arrowheads point from the derived requirement to its source.

«requirement»
Hydroponics monitoring system

id = "R0"
text = "Deliver integrated sensing, local control, outage continuity, BLE commissioning and remote monitoring with traceable validation."

«deriveReqt»

«requirement»
Outage continuity

id = "G3"
text = "Keep the retained measurements and BLE available on battery."

«requirement»
Retained monitoring

id = "R05"
text = "On battery, retain ESP32-S3, BME280, DS18B20, DFR0026, SGP30, and BLE."

«requirement»
External rail availability

id = "R06"
text = "On operating-feed loss, mark pH, TDS and HC-SR04 UNPOWERED; the external-5V relay coil is unavailable."

«requirement»
Safe transition ownership

id = "R07"
text = "Reject external-rail samples across epoch changes. Grant buses only after matching owner-release acknowledgement."

«deriveReqt»

«requirement»
Predictable local behavior

id = "G4"
text = "Bound timing and resource use; keep local decisions independent of communications."

«requirement»
Acquisition cadence

id = "R08"
text = "Use a 1 s frame, 250 ms BME280 sampling and 1 s SGP30 commands in both modes when available."

«requirement»
Local response targets

id = "R09"
text = "Targets under radio load: sample-to-action <= 10 ms; sustained BME280 software detection/action <= 300 ms."

«requirement»
Bounded resource ownership

id = "R10"
text = "Enforce one bus owner, single writers, copied records, finite waits, bounded queues and progress monitoring."

«requirement»
Relay arbitration

id = "R16"
text = "Only Safety writes GPIO15; apply power availability, local interlocks, authorization, expiry and maximum-on time."

«deriveReqt»

«requirement»
Verified delivery

id = "G7"
text = "Deliver the integrated system with traceable validation records."

«requirement»
Integrated validation

id = "R17"
text = "Complete electrical, sensor, calibration, timing, outage and 24 h mixed-mode tests with recorded measurements."

«requirement»
Reproducible release

id = "R18"
text = "Deliver PCB, firmware, protocol, applications and test records with hardware, software, configuration and calibration revision IDs."

Timing figures are acceptance targets. The system-level validation requirement does not assert that any hardware or firmware test has passed.

REQ-03 Satisfaction and verification relationships

req [package Assurance]

«block»
PowerSubsystem
Design responsibility

«satisfy»

«requirement»
Retained monitoring
id = "R05"
text = "On battery, retain ESP32-S3, BME280, DS18B20, DFR0026, SGP30, and BLE."

«verify»

«testCase»
T04 Power-mode and epoch tests
status = specified
evidence = pending

«block»
PowerCoordinator
Design responsibility

«satisfy»

«requirement»
Safe transition ownership
id = "R07"
text = "Reject external-rail samples across epoch changes. Grant buses only after matching owner-release acknowledgement."

«verify»

«testCase»
T04 Power-mode and epoch tests
status = specified
evidence = pending

«block»
Safety
Design responsibility

«satisfy»

«requirement»
Local response targets
id = "R09"
text = "Targets under radio load: sample-to-action <= 10 ms; sustained BME280 software detection/action <= 300 ms."

«verify»

«testCase»
T05 Scheduling and local action
status = specified
evidence = pending

«block»
Configuration
Design responsibility

«satisfy»

«requirement»
Configuration transactions
id = "R12"
text = "Validate, stage, persist a versioned configuration, and apply at a safe boundary; recover the last valid commit."

«verify»

«testCase»
T06 BLE and configuration tests
status = specified
evidence = pending

«block»
RelayChannel
Design responsibility

«satisfy»

«requirement»
Relay arbitration
id = "R16"
text = "Only Safety writes GPIO15; apply power availability, local interlocks, authorization, expiry and maximum-on time."

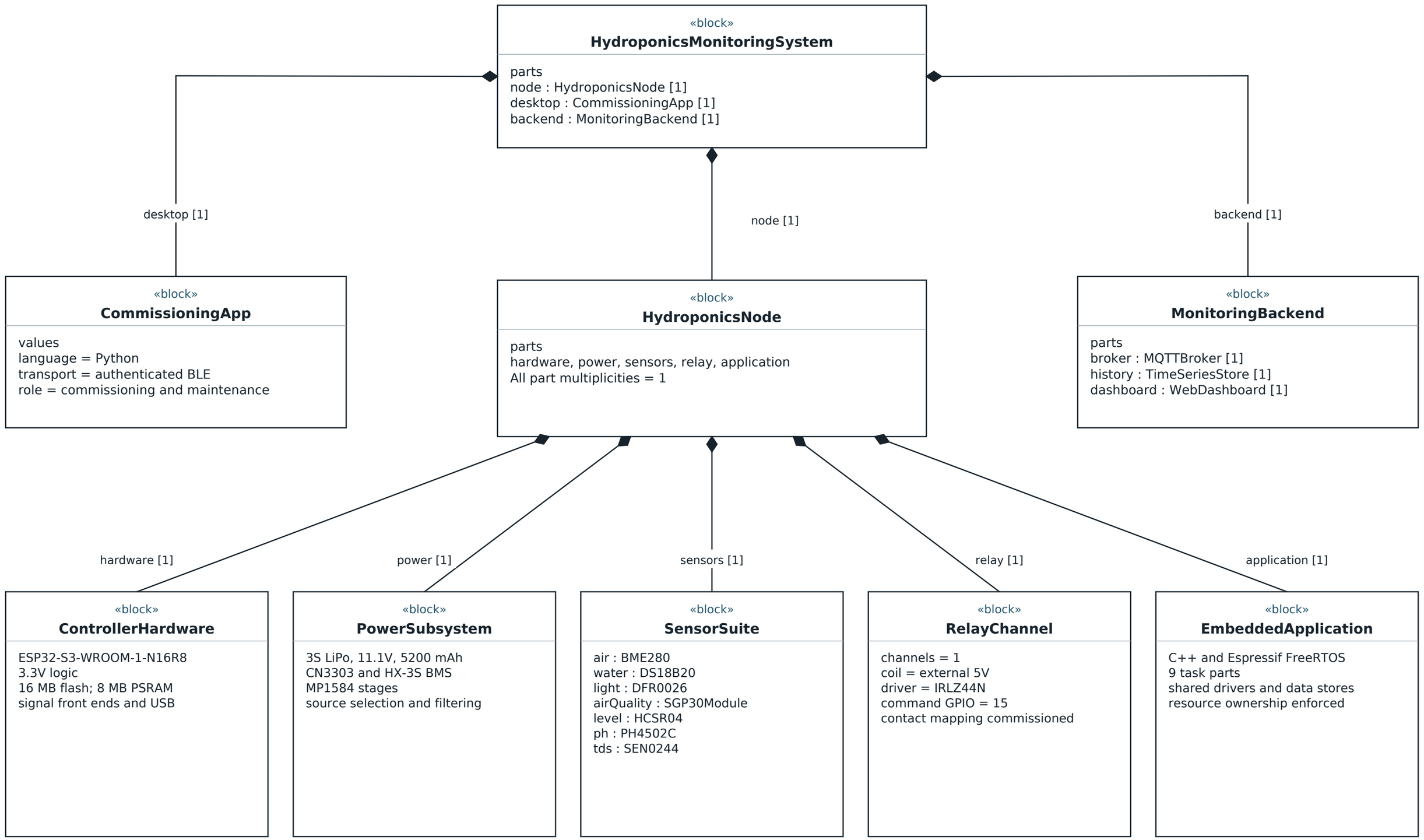
«verify»

«testCase»
T05 Scheduling and local action
status = specified
evidence = pending

Both dependency arrowheads terminate at the requirement. «satisfy» records design responsibility; «verify» identifies a planned test case. All execution evidence remains pending.

BDD-01 System composition

bdd [block HydroponicsMonitoringSystem]



Filled diamonds are at the composite whole. Every shown part has multiplicity 1; SensorSuite contains seven named sensor parts. The charger, measured environment, laptop platform, and controlled load are external context.

BDD-02 Embedded task definitions and ownership

bdd [block EmbeddedApplication]

«block»
AcquisitionWorker {abstract}

operations
serviceDueJob()
quiesce(generation)

«block»
EmbeddedApplication

task parts shown below
other parts: drivers, measurements, nvs
allocation target: ControllerHardware

safety [1]

«block»
Safety

core = 0; priority = 6
sole writer: GPIO15
alarms, interlocks, timers

normal [1]

«block»
AcquisitionNormal

core = 1; priority = 4
all qualified sensors
active only with normal lease
generalization: AcquisitionWorker

network [1]

«block»
WiFiMqtt

core = 0; priority = 2
sole MQTT client owner
bounded network state machine

powerCoordinator [1]

«block»
PowerCoordinator

core = 0; priority = 5
power mode and rail epoch
sole acquisition lease grantor

lowPower [1]

«block»
AcquisitionLowPower

core = 0; priority = 4
four retained sensors
active only with low-power lease
generalization: AcquisitionWorker

BLE [1]

«block»
BLETransport

core = 0; priority = 2
sole application GATT sender
bounded RX/TX and session checks

configuration [1]

«block»
Configuration

core = 0; priority = 2
sole NVS and registry writer
validated, staged configuration

supervisor [1]

«block»
Supervisor

core = 0; priority = 1
progress and memory monitoring
watchdog membership and reset cause

logger [1]

«block»
SerialLogger

core = 1; priority = 1
bounded serial output
blocked in low-power mode

This view selects the nine task parts; driver and data-store parts remain in the canonical model. The two AcquisitionWorker specializations share an engine and are mutually exclusive. Priorities are initial design values.

BDD-03 Interface and data contracts

bdd [package Interfaces and Data]

«interfaceBlock»
DCPower

flow properties
out supply : DCSupply
DCSupply includes voltage and return
Power5V, Power3V3, PackDCFeed, SelectedPower
PackPower has inout supply for the pack link.

«interfaceBlock»
I2CBus

flow properties
inout commands : I2CFrame
inout measurements : I2CFrame
SDA = GPIO21; SCL = GPIO47
BME280 0x76; SGP30 0x58

«interfaceBlock»
OneWireBus

flow properties
inout command_and_data : OneWireFrame
DS18B20 DQ = GPIO14
externally powered at 3.3V

«interfaceBlock»
Analog3V3

flow properties
out level : ADCInputVoltage
upstream AnalogRaw carries Voltage
front ends establish the fitted ADC range

«interfaceBlock»
Digital3V3

flow properties
out level : Logic3V3
TRIG, conditioned ECHO, relay command
and conditioned external-power presence

«interfaceBlock»
USBBus

flow properties
inout data : USBPacket
GPIO19 D−; GPIO20 D+
22 Ω series resistors

«block»
ReadingRecord

value properties
sensorId; metricId; value; unit
acquisitionStart; acquisitionEnd; lastGoodTime
sampleSequence; externalRailEpoch
configurationRevision; calibrationRevision
quality : ReadingQuality
raw ADC value retained where applicable

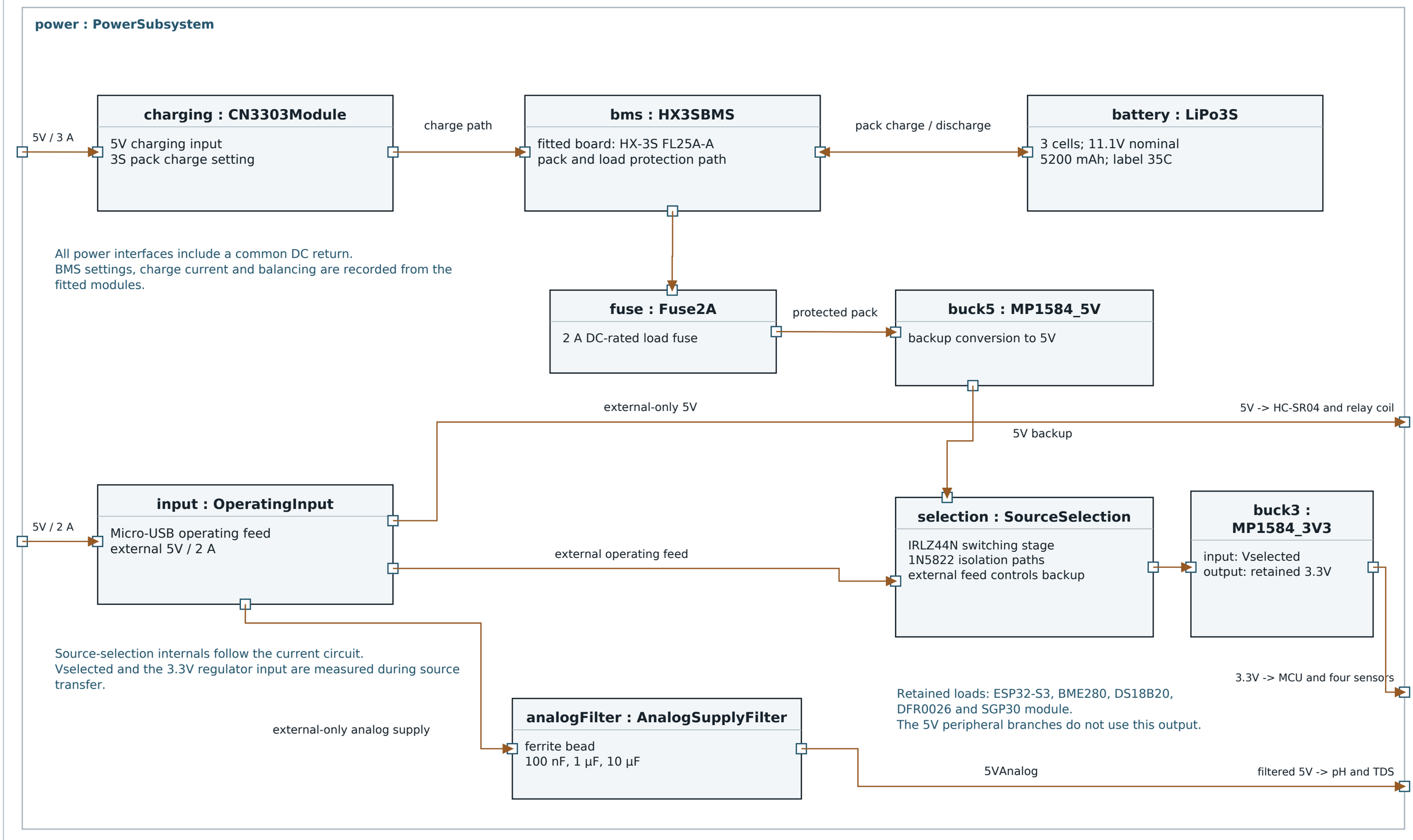
«block»
ControlRequest

value properties
requestId; operation; bounded payload
authenticatedIdentity; sessionGeneration
expectedConfigurationRevision; expiry
result : INVALID | BUSY | REJECTED | APPLIED | FAILED
Credentials are write-only and excluded from logs.

These interface blocks type the logical proxy ports in the IBDs. Unidirectional consumers use conjugated (~) types; inout buses support requests and responses. Supply bundles include positive and return conductors. Record types describe copied data, not pointers shared between tasks.

IBD-01 Power distribution and retained supplies

ibd [block PowerSubsystem]



Proxy ports are typed logical power bundles including a common return. Orange item-flow arrows indicate delivered power; the pack connection is bidirectional. Nominal rail labels do not include path losses. Exact terminal wiring remains governed by the schematic.

IBD-02 Sensor signals and fixed GPIO interfaces

ibd [block HydroponicsNode]

node : HydroponicsNode | signal projection

air : BME280
0x76; air temperature, RH, pressure

airQuality : SGP30Module
0x58; TVOC and CO2 equivalent

I2C bus

water : DS18B20
externally powered 1-Wire

1-Wire; 4.7 kΩ pull-up

light : DFR0026
analog light level

lightRC : LightFrontEnd
10 kΩ
1 µF

usb : USBInterface
Micro-USB D− / D+
22 Ω series on each signal

USB D− / D+

mcu : ESP32S3
ESP32-S3-WROOM-1-N16R8
3.3V logic; 16 MB flash; 8 MB PSRAM

SDA 21 / SCL 47
DQ 14
GPIO1 / ADC1_CH0
GPIO19 / GPIO20
GPIO6 / ADC1_CH5
GPIO7 / ADC1_CH6
GPIO17 TRIG
GPIO18 ECHO
GPIO15 relay
GPIO8 power present

No 5V sensor output is routed directly to an ESP32-S3 input.
Power availability and epoch checks remain firmware responsibilities.

tdsRC : TDSFrontEnd
100 nF
ADC input filtering

tds : SEN0244
external filtered 5V
analog TDS estimate

phRC : PHFrontEnd
10 kΩ / 15 kΩ
1 µF to ground

ph : PH4502C
external filtered 5V
PO used; DO and TO unused

TRIG

level : HCSR04
external 5V
TRIG / ECHO timing

echoRC : EchoFrontEnd
1 kΩ / 2 kΩ divider

ECHO

GPIO15 command

relay : RelayChannel
IRLZ44N; external 5V coil
220 Ω gate; 10 kΩ pull-down

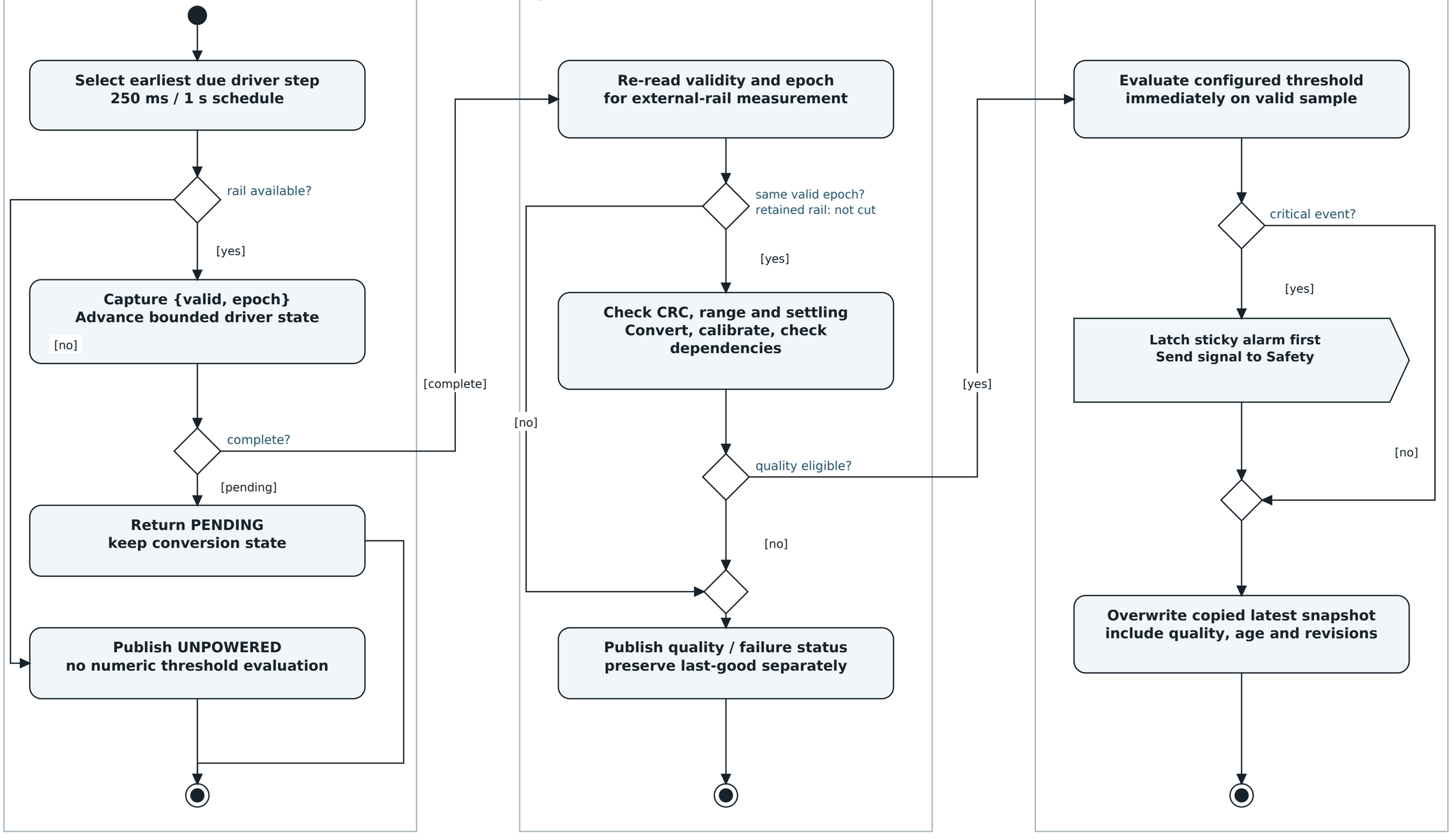
senseRC : PowerSenseFrontEnd
220 Ω / (220 Ω + 220 Ω)
nominal 3.33V from 5.00V

external 5V presence

This is a signal-focused projection: supply connectors are defined in IBD-01. Square proxy ports expose logical interface types. Left sensors use retained 3.3V; right sensors and the relay coil use external 5V. GPIO35-37 remain reserved for PSRAM.

ACT-01 Service one due sensor job

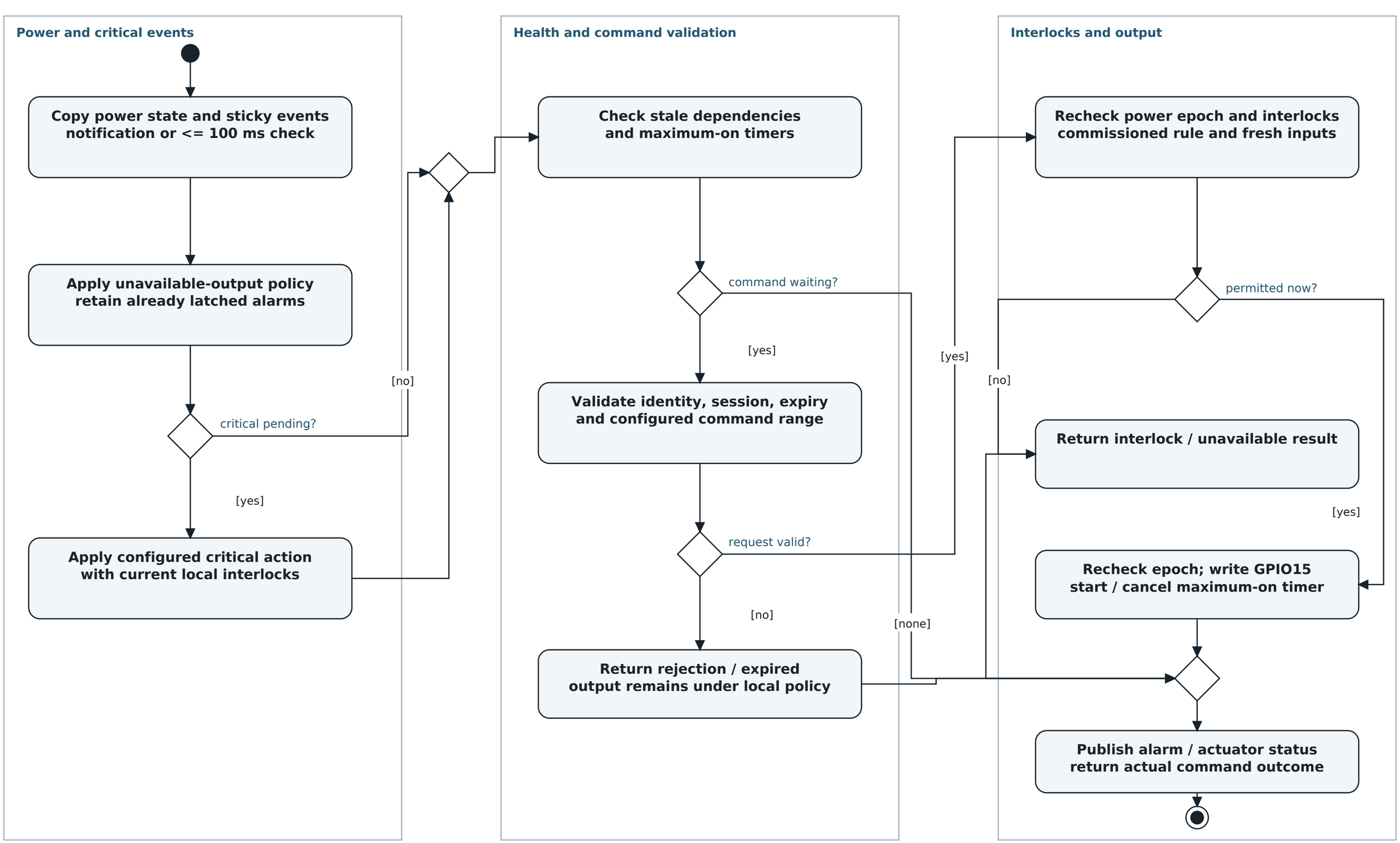
act [activity AcquisitionWorker::serviceDueJob]



Precondition: this worker holds the exclusive bus lease. Partitions show responsibilities executed by that same worker. Each activity-final returns to the scheduler; PENDING retains driver state. The send-signal action does not wait for Safety or a transport.

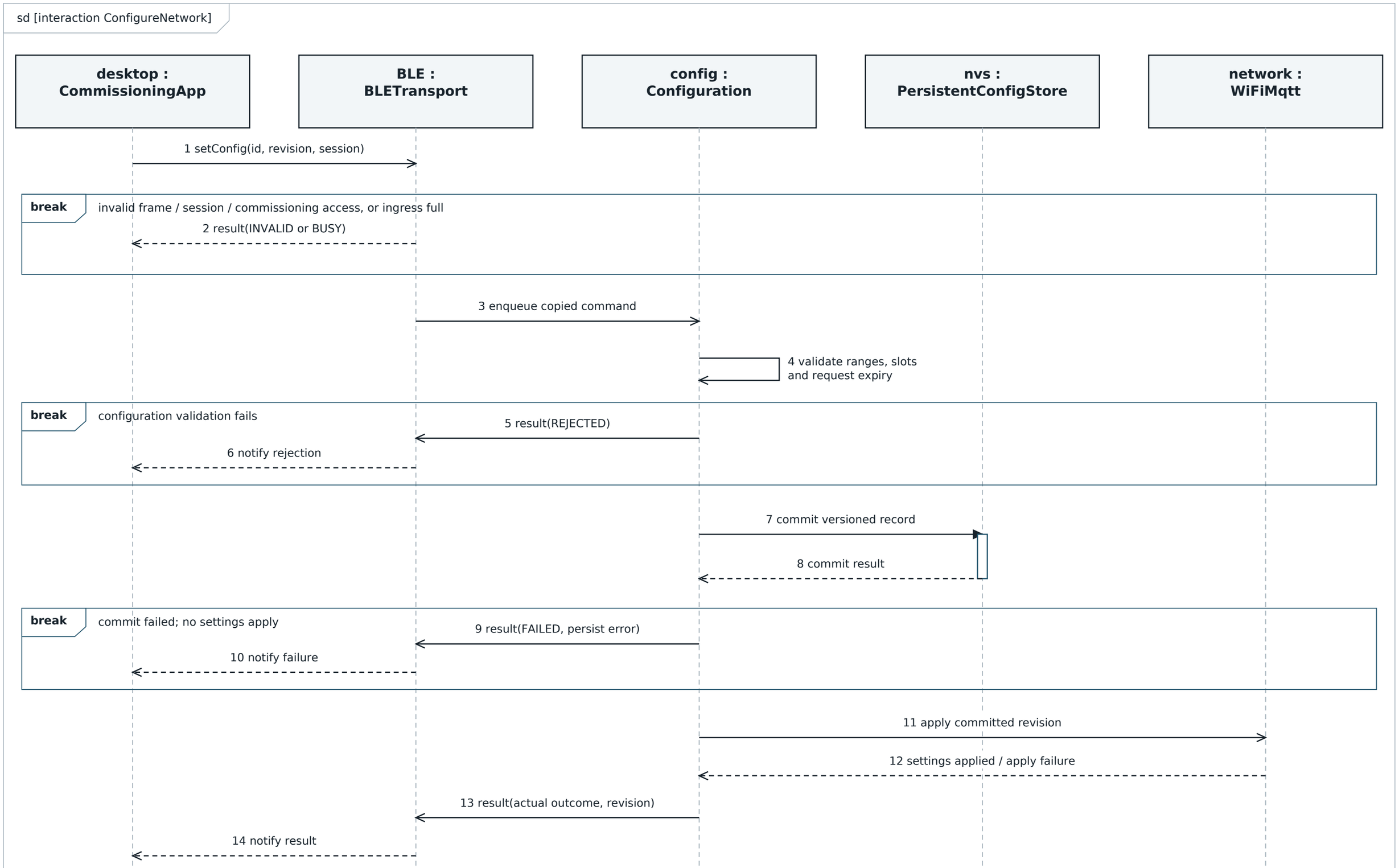
ACT-02 Process a local alarm or control wake

act [activity Safety::processWake]



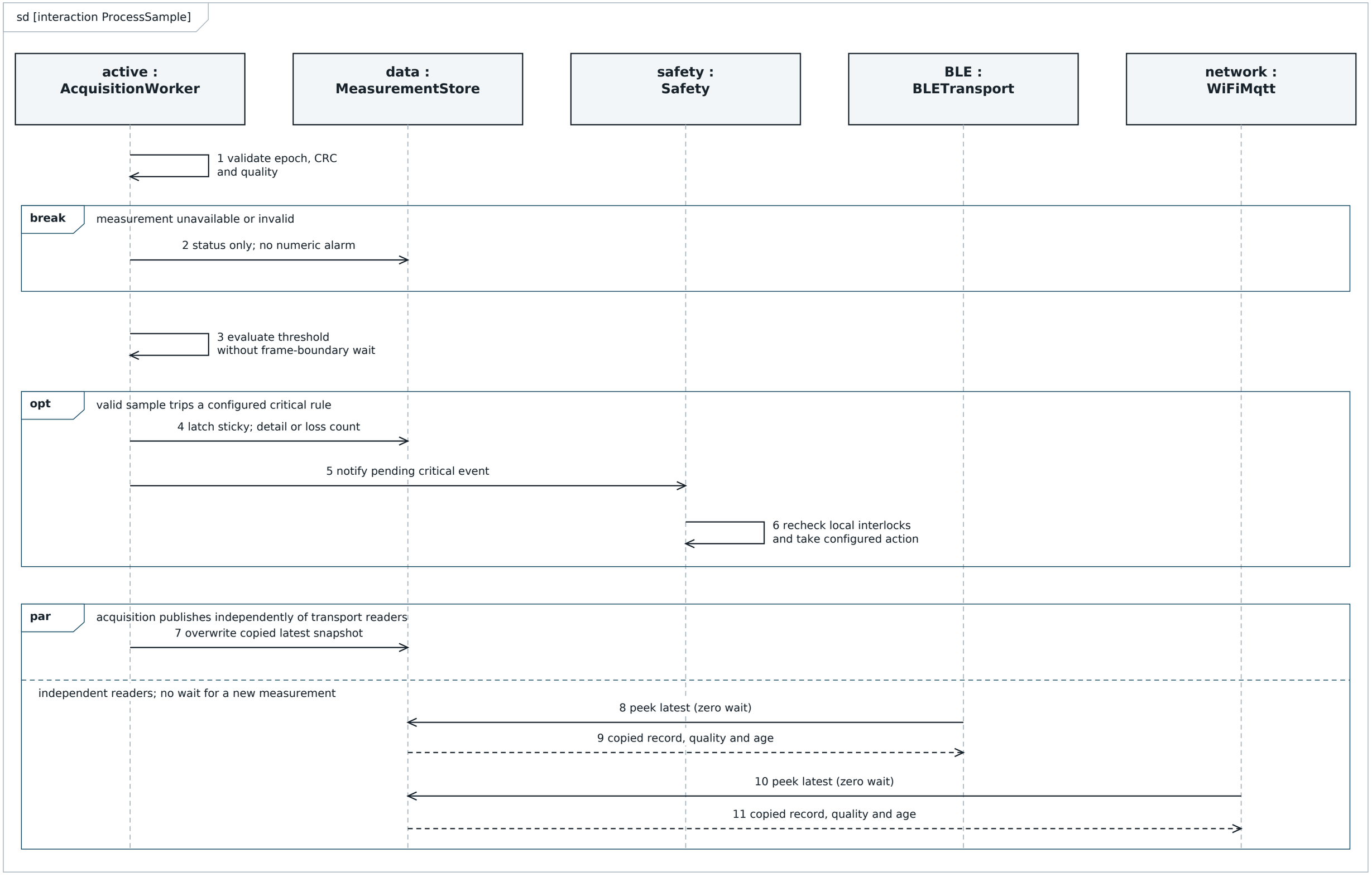
Safety is the sole GPIO15 writer. Critical pending state is handled before commands. A full detail queue cannot erase the sticky alarm. The current rail epoch and interlocks are checked again immediately before actuation; a remote request cannot override them.

SD-01 BLE network configuration transaction



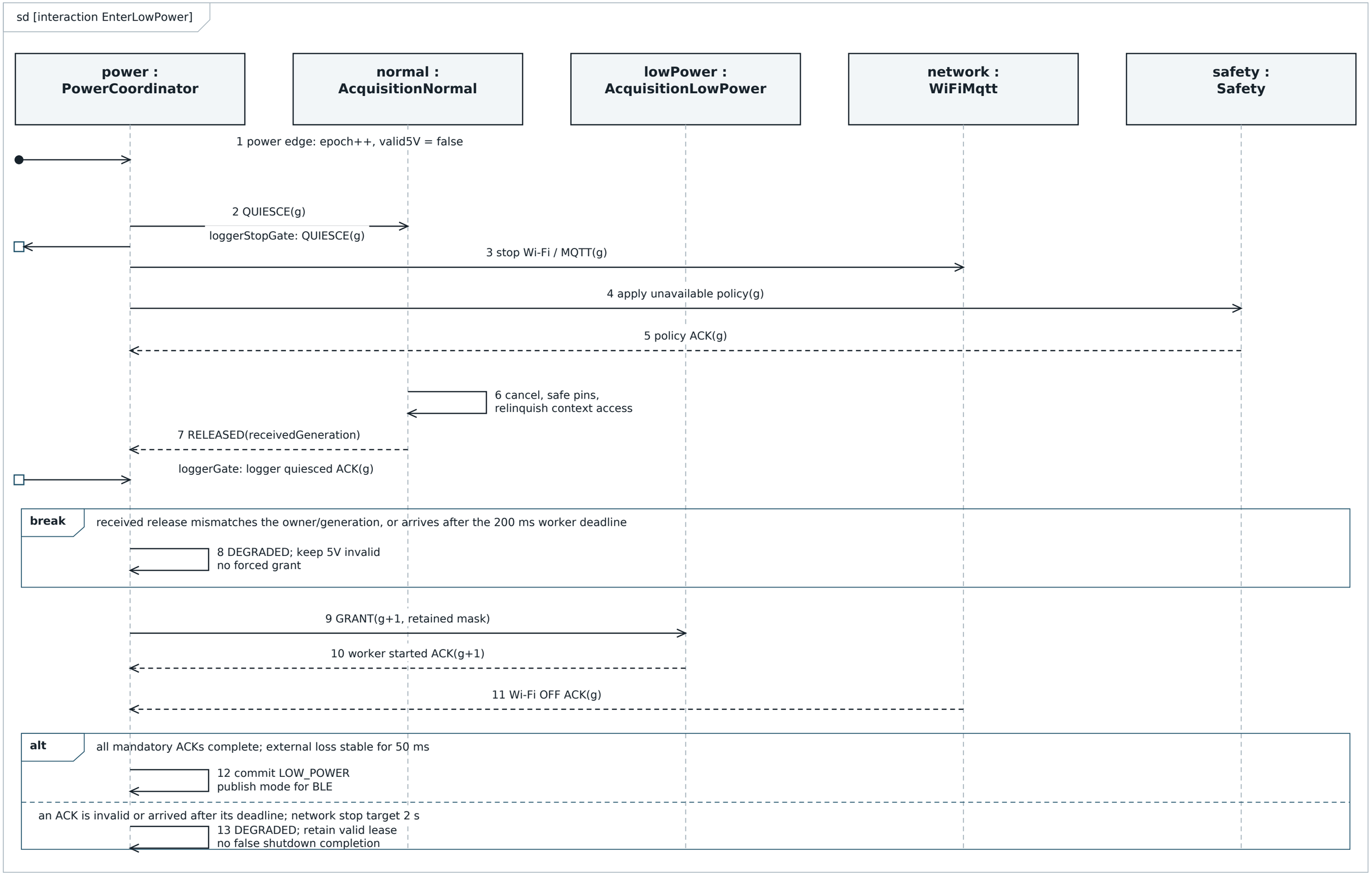
The request is asynchronous at ingress. Each break fragment aborts the rest of this interaction when its guard is true. APPLIED means that network settings were applied; it does not imply a broker connection. Driver-context changes require the quiesce/release protocol before apply.

SD-02 Valid sample and immediate local alarm handling



Local alarm handling precedes network delivery. Sticky alarm state survives detail-queue saturation. In the par fragment acquisition and zero-wait consumers execute independently; a consumer may receive the previous snapshot with its true age. Timing values are targets, not measured durations.

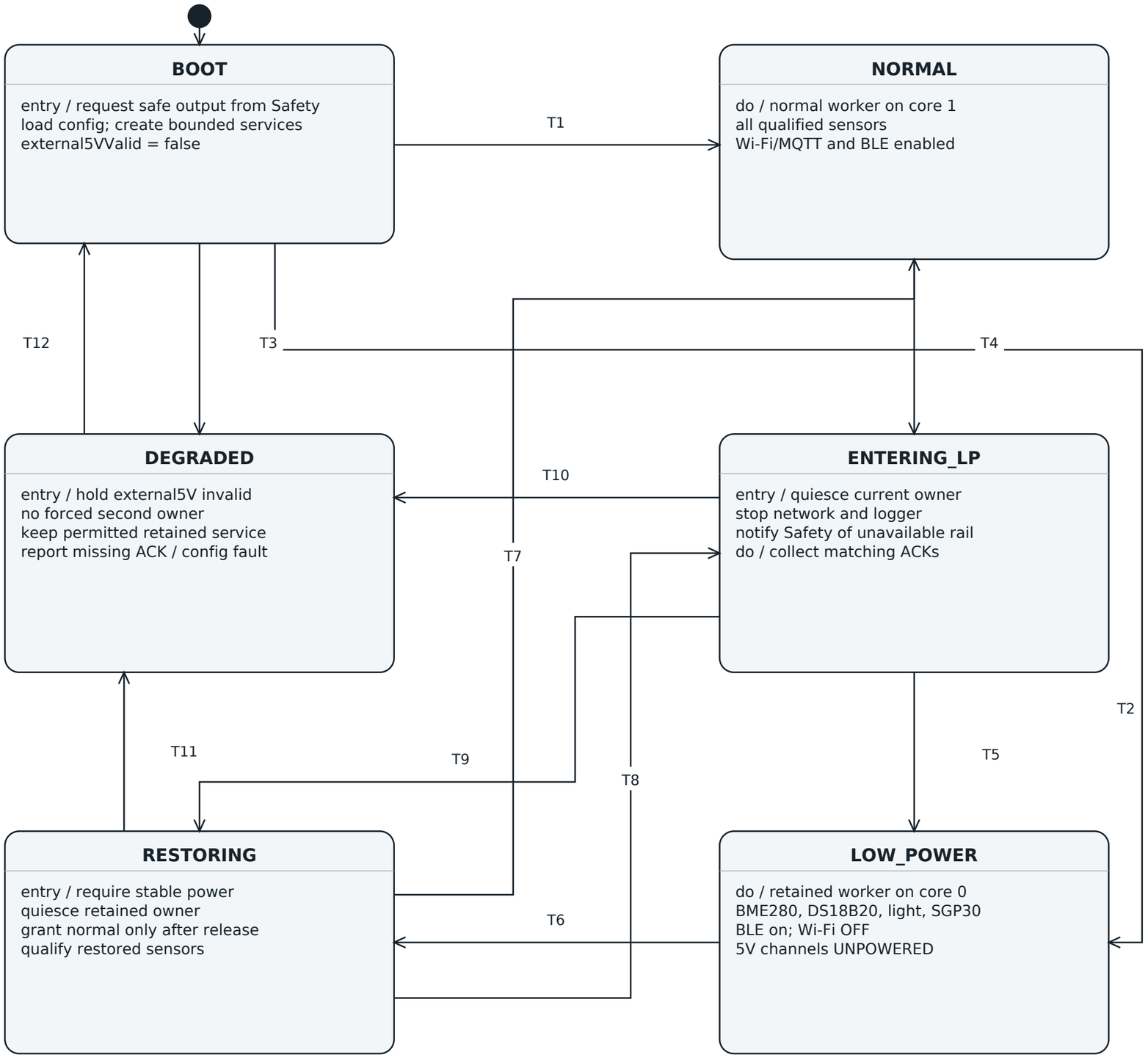
SD-03 External power loss and acquisition handover



This trace shows received-ACK outcomes; missing-ACK deadlines are STM-01 T10/T11. Epoch invalidation precedes debounce. GRANT(g+1) requires matching RELEASED(g). Retained acquisition may start before network shutdown completes, but LOW_POWER is committed only after timely mandatory ACKs.

STM-01 Power operating modes and recovery

stm [state machine PowerCoordinator]



Transition contract

T1 bootComplete [cfgOK && present]
/ Initialize normal lease; start normal services.

T2 bootComplete [cfgOK && absent]
/ Initialize retained lease; Wi-Fi OFF; 5V unavailable.

T3 bootComplete [!cfgOK]
/ Conservative retained-only policy; report not ready.

T4 powerLost [observed edge]
/ ISR increments epoch and invalidates 5V; request quiescence.

T5 transitionReady [absent50ms && mandatoryACKs && retainedLease]
/ Commit LOW_POWER; publish state.

T6 powerPresent [observed edge]
/ Keep 5V invalid; begin stability and rail checks.

T7 restoreComplete [present500ms && railsSettled && normalLease && requiredACKs]
/ Enable external-rail validity; reconnect network asynchronously.

T8 powerLost [observed edge]
/ Abort restore; increment epoch; quiesce current owner.

T9 powerPresent [observed edge]
/ Reconcile ownership; require stable power before normal grant.

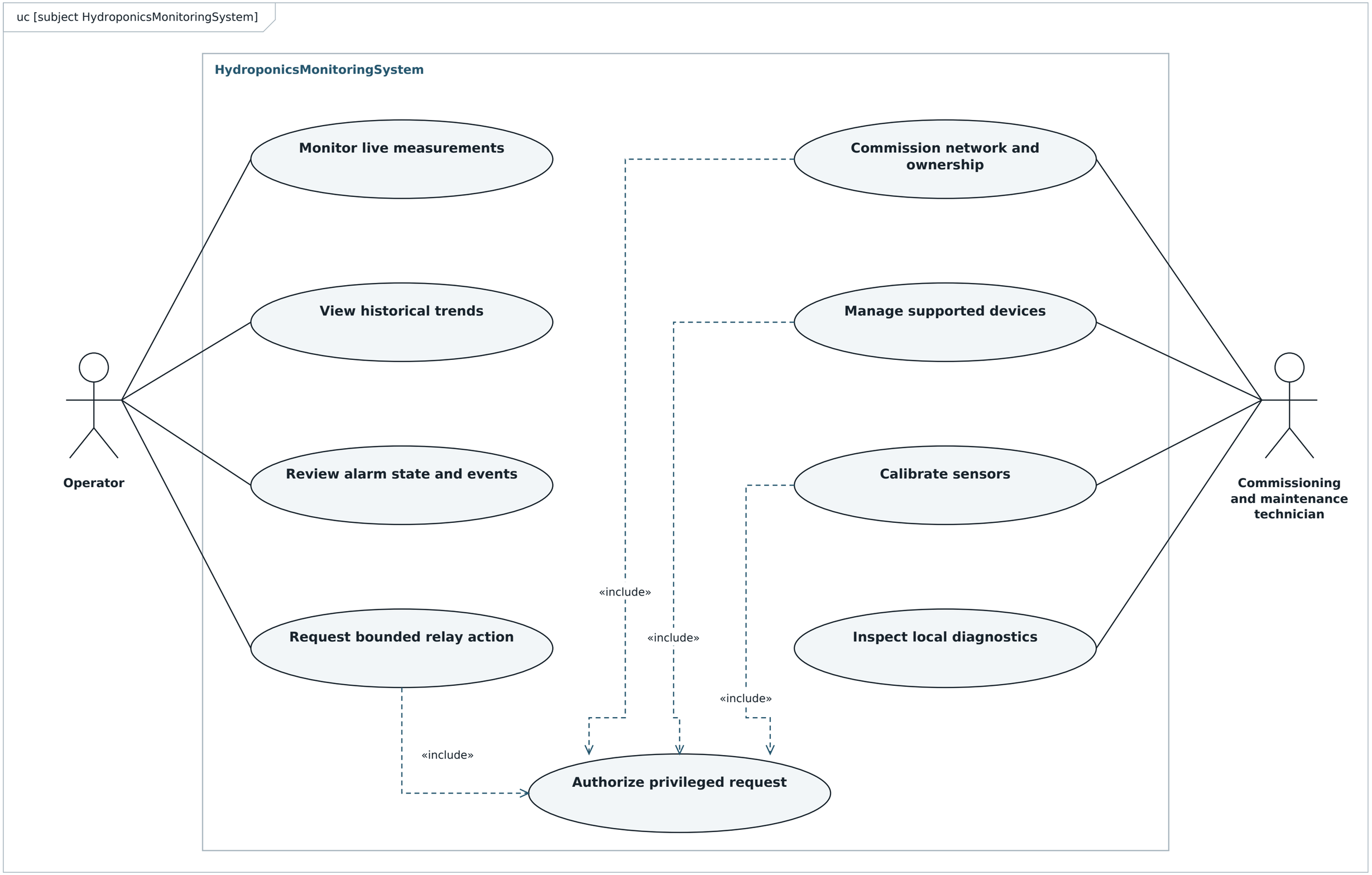
T10 ackDeadline [requiredACKMissing]
/ Keep 5V invalid; never force a new owner.

T11 ackDeadline [requiredACKMissing]
/ Hold invalid rail; preserve exclusive ownership.

T12 recoveryValidated [allOwnershipReconciled && cfgOK]
/ Reinitialize safely with no outstanding owner; re-evaluate power.

Transition IDs on the graph resolve to the complete trigger / guard / effect table. cfgOK includes validated configuration and known power sense. Ownership handover is always quiesce -> matching release -> grant. There is no final state because the controller is reactive.

UC-01 Operational and maintenance use cases



Actors remain outside the whole-system boundary, which contains the field node, desktop application and backend. «include» arrows point to the shared authorization behavior. Actor responsibilities do not replace the commissioned permission policy. Remote history requires backend access; local BLE remains available on battery.

Requirements traceability

«satisfy» and «verify» relationships for every implementation requirement

ID and goal	Requirement	Design responsibility	Planned verification
R01 / G1	Controller implementation	ControllerHardware, PowerSubsystem	T01
R02 / G1	Board interface contract	ControllerHardware	T01
R03 / G2	Seven sensor functions	SensorSuite, SensorDriverSet	T03
R04 / G2	Measurement qualification	SensorDriverSet, MeasurementStore	T03, T09
R05 / G3	Retained monitoring	PowerSubsystem, PowerCoordinator, BLETransport	T02, T04
R06 / G3	External rail availability	PowerSubsystem, PowerCoordinator, Safety	T02, T04
R07 / G3	Safe transition ownership	PowerCoordinator, AcquisitionWorker	T04
R08 / G4	Acquisition cadence	AcquisitionWorker, SensorDriverSet	T03, T05
R09 / G4	Local response targets	AcquisitionWorker, Safety	T05
R10 / G4	Bounded resource ownership	EmbeddedApplication, MeasurementStore, Supervisor	T07, T08
R11 / G5	Secure BLE access	CommissioningApp, BLETransport	T06
R12 / G5	Configuration transactions	Configuration, PersistentConfigStore	T06
R13 / G5	Supported device management	Configuration, CommissioningApp	T06
R14 / G6	Dashboard information	MonitoringBackend, WebDashboard	T07
R15 / G6	Network fault tolerance	WiFiMqtt, BLETransport, MeasurementStore	T07
R16 / G4	Relay arbitration	Safety, RelayChannel	T04, T05, T06
R17 / G7	Integrated validation	HydroponicsMonitoringSystem	T01, T02, T03, T04, T05, T06, T07, T08, T09
R18 / G7	Reproducible release	HydroponicsMonitoringSystem	T10

The links express intended responsibility and verification coverage. A planned test link is not execution evidence. Full requirement text appears in REQ-01/02; the source archive also provides a traceability CSV with diagram coverage.

Planned verification cases

Execution status for T01 to T10 is pending

Test case	Required evidence
T01 PCB and pin inspection	Inspect schematic/layout and continuity against the fixed board profile.
T02 Electrical power validation	Measure charge behavior, rails, transfer, ripple, thermal rise and protection.
T03 Sensor integration	Check identity, cadence, startup, timeout, disconnect and reference response.
T04 Power-mode and epoch tests	Inject loss/return/bounce at driver steps; test ownership, availability and stale ACK rejection.
T05 Scheduling and local action	Trace frame jitter and sample-to-action/detection targets under concurrent Wi-Fi/BLE traffic.
T06 BLE and configuration tests	Test access control, framing, invalid commands, persistence, safe apply and relay authorization.
T07 Network and queue overload	Remove AP/broker for 30 min; congest transports and queues; inspect bounds and local operation.
T08 Fault and endurance tests	Inject stalls; perform 24 h mixed-mode run; record reset, memory, timing and current.
T09 Calibration evidence	Check pH buffers, TDS reference/temperature, tank geometry and light calibration if lux is reported.
T10 Release record audit	Inspect delivered files, revision identifiers, dependencies, calibration and configuration records.

Timing and operating targets

The acquisition frame is 1 s, with BME280 checks every 250 ms and SGP30 commands every 1 s. Initial targets include ≤ 5 ms frame-release jitter, ≤ 10 ms accepted-sample-to-action, and ≤ 300 ms sustained BME280 software detection/action under concurrent radio traffic. Thermal response and actuator mechanics are measured separately.

Power-transition targets start at 50 ms loss debounce, 500 ms stable return, 200 ms worker acknowledgement, and 2 s network shutdown. Hardware measurements determine the final settings. Battery runtime is measured at the complete-board level.

Internal interface register

Logical proxy-port contracts used by IBD-01 and IBD-02

Interface	Flow contract	Application
DC power contracts	Power5V; Power3V3; PackDCFeed; PackPower; SelectedPower	Supply and common return. PackPower is bidirectional; other contracts have producer/consumer roles. SelectedPower retains path losses.
I2CBus	inout commands and measurements	Shared GPIO21 SDA / GPIO47 SCL; BME280 0x76 and SGP30 module 0x58. 4.7 kΩ pull-ups to 3.3V.
OneWireBus	inout command_and_data	DS18B20 on GPIO14, externally powered at 3.3V, with a 4.7 kΩ pull-up.
AnalogRaw	out level : Voltage	Sensor/module output before the fitted signal front end.
Analog3V3	out level : ADCInputVoltage	Conditioned analog signal for ADC1: GPIO1 light, GPIO6 TDS, GPIO7 pH. Actual range and calibration are verified.
Digital3V3	out level : Logic3V3	GPIO17 TRIG, conditioned GPIO18 ECHO, GPIO15 relay command, and conditioned GPIO8 external-power presence.
Echo5V	out pulse : Logic5V	HC-SR04 ECHO before the 1 kΩ / 2 kΩ divider.
USBBus	inout data : USBPacket	GPIO19 D− / GPIO20 D+, with 22 Ω series resistance on each signal.

Direction and delegation

For a unidirectional interface, the producer uses the declared outward flow and the consumer uses the conjugated (~) contract. Assembly connectors pair compatible ends. Delegation connects a boundary port to an internal part port with the same external role. The I2C, 1-Wire and USB buses allow bidirectional exchange.

The machine-readable connector register resolves every endpoint to a declared block port. The power view and signal view are complementary projections; a supply omitted from the signal view remains defined by the rail topology.

Model integrity review

Automated review of the shared model and drawing geometry

The reviewed model contains 26 requirements, including seven goals and 18 implementation requirements; 49 block classifiers; 14 interface contracts; and 30 typed internal connections. Every requested diagram type is represented.

Check category	Passed	Failed
Requirements	100	0
Composition	100	0
Hardware	9	0
Ownership	6	0
Diagram references	333	0
Geometry	1987	0
Notation	801	0
Interfaces	193	0
Activity semantics	67	0
Sequence contracts	33	0
State semantics	33	0
Use cases	25	0
Bounded protocol review	46	0

Bounded behavior review

The abstract lease protocol was explored over generations 0 to 3: 47 reachable states and 470 event steps. No reachable state had two owners; stale acknowledgements changed no ownership, and timeout created no new owner. An injected unsafe-grant variant produced two owners and was detected by the same review harness.

The epoch review evaluated 40 non-wrapping combinations of start/end validity and counter values. Only 4 cases were accepted, all with a valid unchanged epoch. A loss-and-return case with both boolean checks true was rejected by the epoch condition.

What the result establishes

The result establishes model references, coverage, connector compatibility, guarded control flow, explicit merges, sequence ordering, and the stated bounded protocol properties. Drawing bounds, node overlap and connectors crossing unrelated nodes are checked, followed by visual page inspection. It does not execute firmware, measure electrical behavior, or certify native SysML/XMI-tool conformance.

Model decisions and design references

Interpretations needed to keep the views consistent

MD01 The system boundary includes one field node, the Python commissioning application and the monitoring backend. External power, the measured environment and the controlled load remain outside.

MD02 Unknown power sensing or invalid configuration is represented by DEGRADED with conservative retained-only policy. It does not report a completed low-power transition.

MD03 External 5V validity is invalidated on an observed edge before software debounce; restored channels qualify individually. Normal sensing does not wait for MQTT connection or every sensor warm-up.

MD04 Proxy-port types describe logical power and signal bundles. The IBDs preserve the selected topology and do not replace the electrical schematic or establish module performance.

MD05 Generation-tagged release is required before any new acquisition grant. DEGRADED permits no forced transfer from an unacknowledged owner.

MD06 Diagram relationships express design intent. A satisfy or verify dependency does not assert a completed hardware test. No OMG/XMI tool-conformance certification is claimed.

MD07 A DEGRADED recovery returns through BOOT only after ownership is reconciled and configuration/sense are valid. This formalizes a bounded recovery path without granting a second owner.

Commissioning inputs

The fitted module settings, actual rail and source-transfer behavior, sensor calibration, real alarm limits, relay load/contact mapping, safe output policy, build configuration and measured timing remain release records. The current source-selection circuit and both MP1584 stages are retained for experimental validation. Hardware hysteresis remains a later iteration.

Design sources

Redesign Specification Rev 5, sections 1-11; current Pin Layout, page 1; current Power Circuit Diagram, page 1; Firmware Architecture Rev 2, with its older hardware assumptions superseded by the current drawings. Exact source copies and SHA-256 checksums accompany the editable package.

Notation references: OMG SysML 1.6 specification entry and OMG SysML v1 diagram overview. The supplied drawing and JSON formats support this review package; they are not an XMI repository.